Finlay Lemon

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Seeking a 12–16 month internship beginning May–September 2027

Education

Queen's University Sep. 2024 – Apr. 2028
Mechatronics and Robotics Engineering, BASc *Kingston, Ontario*

Experience

QRET Team Member 2024 – Present
QRET, Queen's University *Kingston, Ontario*

- Aiding in design of hybrid-propulsion supersonic sounding rocket.
- Team member since 2024, now participating in CAD design of Avionics stack structural support and access door.

Research Lead, Queen's Rocket Engineering Team Sep. 2025 – Apr. 2026
QRET, Queen's University *Kingston, Ontario*

- Co-led a 15-person team to produce a comprehensive CAN-RGX research report within a 2-month window.
- Investigated high surface-area solids for convection-independent heat exchangers.

Order-handling Specialist May 2025 – Sep. 2025
Coca-Cola Canada Bottling Limited *Kingston, Ontario*

- Assembled and routed over $2,300,000 of product for customers across the Eastern Ontario region.
- Operated powered material-handling equipment in a fast-paced environment with no recorded incidents.

Projects

Lunar Logistics Rover | *SolidWorks, SolidWorks Simulation* June 2026 – Present

- Developing a first-pass design solution to the Canadian Space Agency's Lunar Utility Vehicle (LUV) project.
- Defined electrical, mechanical, and thermal subsystems, maintaining interface control across a rolling development cycle.
- Applied Bekker terramechanics to develop a novel wheel concept (0.45 width x 1.2 m diameter scale) and validated feasibility using FEA, improving flexure FoS to 1.62 from below 1 without significant mass increase.
- Documented the design rationale and trade studies in a standalone technical report.

Room-mapping Rover | *ROS, RViz, Python, SolidWorks* Jan. 2026 – Apr. 2026

- Designed, in a team of 3, an autonomous rover built on the Lynxmotion 4WD platform.
- Integrated ROS with SLAM to enable LiDAR-based mapping and autonomous point-to-point navigation.

Desktop Windtunnel | *SolidWorks, FDM 3D Printing, COTS Integration* Oct. 2025 – Nov. 2025

- Prototyped a functioning streamline-visible windtunnel from 3D-printed and COTS parts, built as a minimum-cost alternative to the large commercial units used at Queen's.
- Used a modular design approach, allowing manufacture in approximately 20 hours with household filament.

CAN-RGX Proposal | *Word, Teams, Zotero* Sep. 2025 – Dec. 2025

- Co-authored a CAN-RGX research project proposal for the 2025 cycle.
- Built the proposal around thermal exchange using TPMS high surface-area volumes in microgravity, targeting heat exchangers that do not rely on convection.
- Delivered the complete submission package on schedule in tight 2-month deadline, coordinating 15 personnel, sourcing, and internal review cycles.

Technical Skills

CAD: SolidWorks (4 years), Siemens NX (6 months)
Software: ROS, RViz, Gazebo, Python, C++, MATLAB / Simulink, Altera FPGA **Analysis**: SolidWorks Simulation (FEA), RViz, Excel
Certifications: Amateur Radio — ISED Canada 'Advanced' Qualification